ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

DISTANCE LEARNING PROGRAMME

(Academic Session : 2024 - 2025)

JEE (Advanced)
MAJOR TEST
16-02-2025

JEE(Main + Advanced) : LEADER TEST SERIES / JOINT PACKAGE COURSE

Test Type : Full Syllabus

ANSWER KEY — PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	C	D	C		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,C,D	A,D	C			
SECTION-II (i)	Q.	1	2	3	4		
	A.	4.15 to 4.20	4.50	40.00	2.00		
SECTION-II (ii)	Q.	5	6	7	8	9	10
	A.	5	6	6	44	6	8

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	D	C	C	B		
SECTION-I (ii)	Q.	5	6	7			
	A.	B,C,D	B,D	A,B,C			
SECTION-II (i)	Q.	1	2	3	4		
	A.	2.00	4.00	4.00	12.00		
SECTION-II (ii)	Q.	5	6	7	8	9	10
	A.	84	4	1	8	4	4

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	D	B	B		
SECTION-I (ii)	Q.	5	6	7			
	A.	B,C	A,B,D	A,C			
SECTION-II (i)	Q.	1	2	3	4		
	A.	7.00	24.00	3360.00	60.00		
SECTION-II (ii)	Q.	5	6	7	8	9	10
	A.	2	0	1540	3	5	2

HINT – SHEET

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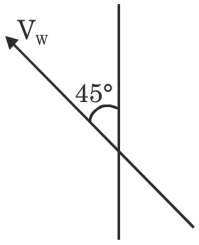
PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A)

$\vec{V}_{AW} = \frac{2r}{t}\hat{i}$

$\vec{V}_{BW} = \frac{\sqrt{2}r}{t}\left(\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right) = \frac{r}{t}\hat{i} + \frac{r}{t}\hat{j}$



Let $\vec{V}_W = x\hat{i} + y\hat{j}$

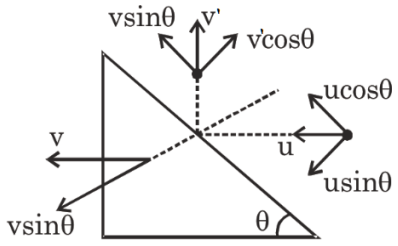
$\frac{2r}{t}\hat{i} = \left(x + \frac{r}{t}\right)\hat{i} + \left(y + \frac{r}{t}\right)\hat{j}$

$x = -\frac{r}{t}$

$y = \frac{r}{t}$

$\theta = 45°$ with verticle

2. Ans (C)



$v\sin\theta - (-v'\cos\theta) = e\,u\sin\theta$

$e = \frac{v\sin\theta + v'\cos\theta}{u\sin\theta} = \frac{v}{u} + \frac{v'}{u}\cot\theta$

From COM $v = \frac{mu}{M}$

& $v'\sin\theta = u\cos\theta$

$\frac{v'}{u} = \cot\theta$

3. Ans (D)

$Q_2 = \frac{(2\omega_1)L}{R}$ (as frequency is doubled) = 2Q,

where $Q_1 = \frac{L\omega_1}{R} \Rightarrow$ Q is doubled

$I_1 = \frac{V}{\sqrt{(L\omega_1)^2 + R^2}} \simeq \frac{V}{L\omega_1}$ (as $L\omega >> R$ for high Q coils) Similarly, $I_2 \simeq \frac{V}{L(2\omega_1)} \Rightarrow I_2 = \frac{I_1}{2}$

$P_2 = I_2R^2 = \left(\frac{I_1}{2}\right)R^2 = \frac{I_2R^2}{4} = \frac{P_1}{4} \Rightarrow$ P decreases 4 times.

4. Ans (C)

Let $T \propto P^a d^b E^c$

Writing dimensions on both sides.

$[M^0L^0T] = [ML^{-1}T^{-2}]a\,[ML^{-3}]b\,[ML^2T^{-2}]c$

$\Rightarrow [M^0L^0T] = M^{a+b+c}L^{-a-3}b^{+2c}T^{-2a-2c}$

Thus $a+b+c=0$, $-a-3b+2c=0$, $-2a-2c=1$

On solving these equations, we get $a = \frac{-5}{6}$, $b = \frac{1}{2}$ and $c = \frac{1}{3}$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (A,C,D)

$f_B = \frac{c}{c+u}f_A < f_A$

Heard by A $f_B' = \frac{c-u}{c}f_B = \frac{c-u}{c+u}f_A < f_A$

6. Ans (A,D)

$Mg\frac{L}{2} = \frac{1}{2}\left(\frac{ML^2}{3}\right)\omega_0^2$

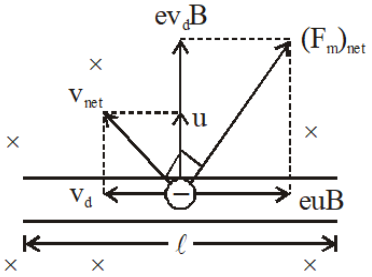
$\omega = \sqrt{\frac{3g}{2L}}$

$e = \frac{v - \omega(L-r)}{(L-r)\omega_0} = 1$

COAM, $\frac{ML^2}{3}\omega_0 = \frac{ML^2}{3}\omega + mv(L-r)$

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7. Ans (C)



$W_{magnetic} = 0$

To maintain the current work has to be done by battery

$W_{batt} = \int (nA\ell)euBv_d dt$

$I = neAv_d = I\ell B \int udt$

$W_{Batt} = I\ell Bh$

$W_{Battery} + W_{Gravity} = \Delta KE$

PART-1 : PHYSICS

SECTION-II (i)

1. Ans (4.15 to 4.20)

$$F = \int_{0.02}^{0.08} \frac{\lambda}{2\pi\varepsilon_0}\frac{\lambda' dx}{x}; \; F = \frac{2\lambda\lambda'}{4\pi\varepsilon_0}\ell n4$$

$$= 2 \times 9 \times 10^9 \times 10^{-7} \times \frac{10 \times 10^{-9}}{0.06} 2\ell n2$$

$$= 2 \times 3 \times 10^9 \times 10^{-7} \times 1000 \times 10^{-9} \ln 2$$

$$= 6 \times 0.693 \times 10^{-7} \times 10^3$$

$$F = 4.2 \times 10^{-4} \text{ N}$$

2. Ans (4.50)

$$F = ma; \; \frac{qE}{m} = a$$

$$a = \frac{10 \times 10^{-9}}{1} \times \frac{2\lambda}{4\pi\varepsilon_0}\frac{1}{r}$$

$$= \frac{10^{-8} \times 2 \times 9 \times 10^9 \times 10^{-7}}{0.04}$$

$$a = 45.0 \times 10^{-6} = 4.5 \times 10^{-4} \text{ m/s}^2$$

3. Ans (40.00)

$$Q = \frac{h^2}{24m\lambda^2} = K_B + K_C$$

where k is the kinetic energy

From conservation of linear momentum

$$|\vec{P}_C| = |\vec{P}_B| = P$$

$$\because K = \frac{P^2}{2m} \therefore Q = \frac{P^2}{2 \times 12m} + \frac{P^2}{2 \times 4m}$$

$$\frac{h^2}{24m\lambda^2} = \frac{P^2}{2}\left(\frac{1}{12m} + \frac{1}{4m}\right)$$

$$\frac{h^2}{12m\lambda^2} = P^2\left(\frac{1+3}{12m}\right) = \frac{P^2}{3m}$$

$$\Rightarrow P^2 = \frac{h^2 \times 3m}{12m\lambda^2} \Rightarrow P = \frac{h}{2\lambda}$$

$$\Rightarrow \frac{h}{P} = 2\lambda = \text{de Broglie wavelength}$$

$$\because |\vec{P}_B| = |\vec{P}_C|$$

$$\text{Mass defect, } \Delta m = \frac{Q}{C^2} = \frac{h^2}{24mc^2\lambda^2}$$

$$\therefore m_A = 16m + \frac{h^2}{24mc^2\lambda^2}$$

PART-1 : PHYSICS

SECTION-II (ii)

5. Ans (5)

for motion in medium $\sin\theta_c = \frac{4}{5}$

$\therefore \cos\theta_c = \frac{3}{5}$

$w = 2\pi(1 - \cos\theta_c) = 2\pi\left(1 - \frac{3}{5}\right) = \frac{4\pi}{5}$

$\therefore$ % of light coming out $= \frac{\left(\frac{\omega}{4\pi} \times 100\right)}{2}\% = 10\%$

$\therefore$ % of light travel throug medium = 40 %

Now for motion of source in vacuum half of emitted light will travel through medium i.e. 25%

In one complete revolution % of light through medium = 40+25 = 65%

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6. **Ans (6)**

$W_{gas} + W_{spring} + W_{after} = 0$

$W_{gas} - \frac{1}{2} \times 200 \times 10^3 \times (0.2)^2 - 10^5 \times 0.05 \times 0.4 = 0$

$W_{gas} = 6000\ J = 6\ kJ$

7. **Ans (6)**

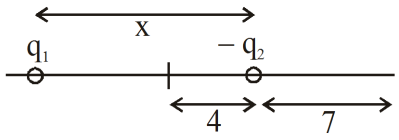
Capillary height $h = \frac{2T\ \cos\theta}{r\rho g}$

$\Rightarrow h = 59.6$ mm

here 59.6 mm is greater than the protruding part hence water will rise in the capillary of insufficient height 25 mm.

Now, $R = \frac{2T}{\rho g h} = 0.6$ mm

8. **Ans (44)**



$\frac{q_1}{x-4} - \frac{q_2}{4} = 0$

$\frac{q_1}{x+7} - \frac{q_2}{7} = 0$

$\frac{q_1}{q_2} = \frac{x-4}{4}$

$\frac{q_1}{q_2} = \frac{x+7}{7}$

$\frac{x-4}{4} = \frac{x+7}{7}$

$7x - 28 = 4x + 28 \quad 3x = 56 \quad x = \frac{56}{3}$

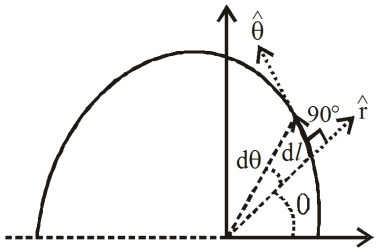
$\frac{q_1}{q_2} = \frac{\frac{56}{3}+7}{7} = \frac{11}{3}$

$|q2| = +12\ \mu c \Rightarrow q_1 = 12 \times \frac{11}{3} = 44\ \mu c$

9. **Ans (6)**

$R = \frac{V}{I} = \frac{10^3}{1.43} = 700\Omega = 7 \times 10^{-1} k\Omega$

10. **Ans (8)**



$d\vec{B} = \frac{\mu_0 I}{4\pi} \frac{\vec{dl} \times \hat{r}}{r^2}$

$dB = \frac{\mu_0 I}{4\pi} \frac{r d\theta}{r^2}$

$B = \frac{\mu_0 I}{4\pi} \int_0^{\pi} \frac{d\theta}{r}$

$B = \frac{\mu_0 I}{8} \ln 3$

PART-2 : CHEMISTRY

SECTION-I (i)

1. **Ans (D)**

Ncert page 444, Chapter 16 and line 20.

Low level of nor-adrenaline is the cause of depression. Anti-depressant drugs are needed to cure this problem.

3. **Ans (C)**

Finally, presure in both flask will be same.

$\therefore n_1 T_1 = n_2 T_2$

4. **Ans (B)**

Electron withdrawing group like (NO_2) increase stability of alkoxide ion by dispersal of negative charge. In (B) and (C) structures negative charge is in conjugation with double bond and also stabilised by electron withdrawing effect of nitro group.

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PART-2 : CHEMISTRY

SECTION-I (ii)

5. Ans (B,C,D)

(A) Due to high activation energy chemisorption initially increases then decreases with rise in temperature.

(B) Solid catalyst adsorb gas, by strong chemical bonds i.e. by chemical adsorption.

(C) The molecules of adsorbate are held on surface of solid adsorbent due to this entropy decreases. i.e. ΔS is negative. (D) Extent of adsorption of gases on charcoal increases with increase in pressure of the gas

6. Ans (B,D)

A : $Cu_2[Fe(CN)_6]$

B : $[Cu(NH_3)_4]^{2+}$

C : $[Fe(CN)_6]^{4-}$

D : $Cu(OH)_2$

7. Ans (A,B,C)

Since the intermediate anion is planar, the stereochemistry of the starting substrate is irrelevant.

PART-2 : CHEMISTRY

SECTION-II (i)

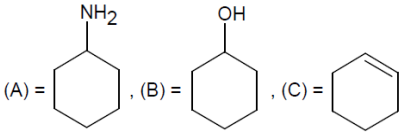
1. Ans (2.00)

Fact

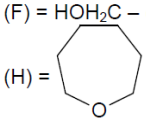
2. Ans (4.00)

Only Ti, V, Cr and Mn show higher valency then Fe

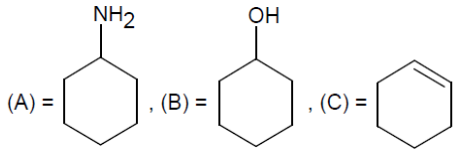
3. Ans (4.00)



(D) = $HOOC-(CH_2)_4-COOH$, (E) = $HOH_2C-(CH_4)_4-CH_2OH$
(F) = $HOH_2C-(CH_2)_4-CH_2Cl$, G = $O^-CH_2-(CH_2)_4-CH_2Cl$



4. Ans (12.00)

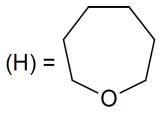


(D) = $HOOC-(CH_2)_4-COOH$,

(E) = $HOH_2C-(CH_4)_4-CH_2OH$

(F) = $HOH_2C-(CH_2)_4-CH_2Cl$,

(G) = $O^-CH_2-(CH_2)_4-CH_2Cl$



PART-2 : CHEMISTRY

SECTION-II (ii)

5. Ans (84)

$X(liq) \longrightarrow X(g)$

$$\Delta S_{300} = \Delta S_{360} + \Delta C_p \ln\frac{300}{360}$$

$$\Delta S_{300} = \frac{28.8\times 1000}{360} + (82-136)$$

$$\ln\frac{300}{360} = 84.32\ JK^{-1}\ mol^{-1}$$

6. Ans (4)

i, ii, v, vi

7. Ans (1)

$$z = 1 + \frac{Pb}{RT}$$

$$\frac{dZ}{dP} = \frac{b}{RT} = \frac{1}{5.6}$$

$$\Rightarrow b = \frac{RT}{5.6} = 4$$

$$\text{Now, } 4\times\left(\frac{4}{3}\pi r^3\times N_A\right) = 4$$

$$\Rightarrow \frac{4}{3}\pi r^3\times N_A = 1$$

8. Ans (8)

It has six geometrical isomers in which two exists as enantiomer pairs, so total eight forms

9. Ans (4)

$$\frac{K_{sp}}{K_a} = \frac{x^2}{[H^{\oplus}]};\ x = \text{solubility}$$

$$x = \sqrt{\frac{K_{sp}}{K_a}\times(H^{\oplus})} = \sqrt{\frac{8\times10^{-10}}{5\times10^{-10}}\times10^{-3}} = 4\times10^{-2}M$$

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10. **Ans (4)**

Canary yellow precipitate is $(NH_4)_3PO_4.12MoO_3$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (A)**

$2x^2 + 6yx + 5y^2 - 1 = 0$

$x \in R \therefore D \geq 0$

$36y^2 - 8(5y^2 - 1) \geq 0$

$8 - 4y^2 \geq 0$

$y^2 \leq 2$

2. **Ans (D)**

$3 - 2k - 2 = 0$

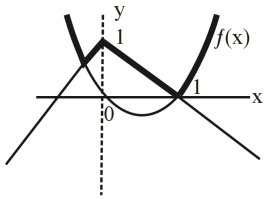
for continuity for differentiability

$f'(1^-) = f'(1^+)$

$1 = -2k$

$k = -\frac{1}{2}$

$\therefore$ No value of k.



3. **Ans (B)**

$\left|\frac{\alpha + \alpha^2 + 3}{\sqrt{2}}\right| = 2\sqrt{2}$

$\alpha + \alpha^2 + 3 = 4$

$\alpha^2 + \alpha - 1 = 0 \Rightarrow \alpha = \frac{-1+\sqrt{5}}{2}$ or $\frac{-1-\sqrt{5}}{2}$

$\therefore$ α has only one value in first quadrant.

4. **Ans (B)**

Assume (h, k) be the middle point of chord

$\therefore$ Its equation is

$hx - ky = h^2 - k^2 \Rightarrow y = \frac{h}{k}x + \frac{k^2 - h^2}{k}$

Now using $c = \frac{a}{m}$, we get

$\frac{k^2 - h^2}{k} = \frac{5}{h}.k$

$\Rightarrow x^3 = y^2(x - 5)$

$\therefore$ a = 5

PART-3 : MATHEMATICS

SECTION-I (ii)

5. **Ans (B,C)**

(A) $g'(x) = f(x)$ ...(1)

tangent to $y = f(x) : y - f(x_1)$

$= f'(x_1)(x - x_1)$ ...(2)

tangent to $y = g(x) : y - g(x_1)$

$= g'(x_1)(x - x_1)$ ...(3)

for (2) and (3), at y = 0

$\frac{f(x_1)}{g(x_1)} = \frac{f'(x_1)}{g'(x_1)}$

$\Rightarrow \frac{f'(x)}{f(x)} = \frac{g'(x)}{g(x)}$

$\Rightarrow \ell nf(x) = \ell n(g(x)) + \ell nc$

$\Rightarrow \frac{f(x)}{g(x)} = c$

put x = 0, we get $c = \frac{1}{4}$

$\Rightarrow f(x) = \frac{g(x)}{4}$

$g'(x) = \frac{g(x)}{4} \Rightarrow g(x) = 4e^{\frac{x}{4}}$

$\Rightarrow f(x) = e^{x/4}$

(B) $I = \int_{-1}^{1} \frac{x^2 + 4}{4 + 4e^{x/4}} dx$

Apply king,

$I = \int_{-1}^{1} \frac{x^2 + 4}{4 + 4e^{-x/4}} dx$

Add $2I = \int_{-1}^{1} \frac{x^2 + 4}{4} dx$

$\Rightarrow I = \frac{13}{12}$

(C) $h(x) = f(x) - g(x)$

$h(x) = e^{x/4} - 4e^{x/4}$

$h'(x) = e^{x/4}\left[\frac{1}{4} - 1\right]$ (–ve)

$\therefore$ h(x) is decreasing

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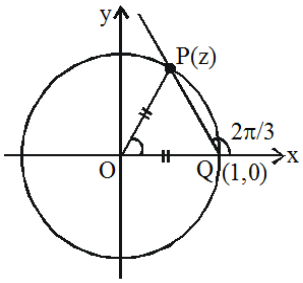
6. **Ans (A,B,D)**

$\because |z| = 1$

$\Rightarrow$ z lies on a circle whose centre is at origin and radius = 1

and $\because \quad \arg(z-1) = \frac{2\pi}{3}$

$\Rightarrow$ z lies on a ray emanating from (1, 0) making an angle of $\frac{2\pi}{3}$ with positive real axis.



$\because \angle PQX = \frac{2\pi}{3}$

$\Rightarrow \angle PQO = \frac{\pi}{3}$

and $\quad OP = OQ$

$\Rightarrow$ triangle OPQ is equilateral

$\Rightarrow OP = OQ = PQ \quad \Rightarrow |z| = |z-1| = 1$

$\Rightarrow$ (D) is correct

and amp(z) = 60°

$\Rightarrow a = \cos60° + i\sin60° = \frac{1+i\sqrt{3}}{2} = -\omega^2$

$\Rightarrow$ (B) is correct

$\Rightarrow$ (C) is correct

Now $z^2 = \omega^4 = \omega \quad \Rightarrow z^2 + z^4 = -1$

$\Rightarrow \text{amp}(z^2 + z^4) = \pi \quad \Rightarrow$ (A) is correct.

7. **Ans (A,C)**

$I + F = \left(15+\sqrt{224}\right)^n = {}^nC_0(15)^n + {}^nC_1(15)^{n-1}\left(\sqrt{224}\right)^1 + \ldots + {}^nC_n(15)^0(\sqrt{224})^n$

Let $F' = \left(15-\sqrt{224}\right)^n$

$= {}^nC_0(15)^n - {}^nC_1(15)^{n-1}(\sqrt{224})^1 + \ldots$

$\Rightarrow \quad I + F + F' = \text{even}$

$\Rightarrow \quad 0 < F + F' < 2 \quad \Rightarrow \quad F + F' = 1$

$\Rightarrow \quad F' = 1 - F$

$\Rightarrow \quad I = \text{odd and } x(1-F)$

$= (15+\sqrt{224})^n(15-\sqrt{224})^n = 1$

PART-3 : MATHEMATICS

SECTION-II (i)

1. **Ans (7.00)**

Normal : (Given)

$4x - 3y = 7 \qquad \ldots. (1)$

Ellipse : $\frac{x^2}{32} + \frac{y^2}{18} = 1$

let point P : (x_1, y_1)

$\therefore$ Normal : $\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$

$\Rightarrow \frac{32x}{x_1} - \frac{18y}{y_1} = 14$

$\Rightarrow (32y_1)x - (18x_1)y = 14x_1y_1 \qquad \ldots. (2)$

compare (1) & (2)

$\frac{32y_1}{4} = \frac{-18x_1}{-3} = \frac{14x_1y_1}{7}$

$\Rightarrow 8y_1 = 6x_1 = 2x_1y_1$

$\Rightarrow (x_1, y_1) \equiv (4, 3)$ Ans.

2. **Ans (24.00)**

Area of ellipse = π ab = $\pi\sqrt{32 \times 18} = 24\pi$

3. **Ans (3360.00)**

The order of vowels does not change means, first u occurs, then i and then e must occur. The total number of arrangements is $\frac{8!}{2!}$ (since there are two l)

In each of these arrangements, vowels may occur in 3! ways. Therefore the number of arrangements in which u, i, e occurs in this order is :

$\frac{8!}{2!} \times \frac{1}{3!} = 3360$

4. **Ans (60.00)**

Keeping u, i, e in their respective places, the number of arrangements is

$\frac{5!}{2!} = 60$

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PART-3 : MATHEMATICS

SECTION-II (ii)

5. Ans (2)

Differentiate given function

$3f^2(x) \cdot f'(x) = x f^2(x)$

$f^2(x) \cdot [3f'(x) - x] = 0$

$\because f^2(x) > 0 \quad (\because f(0) = 0 \ \& \ f \text{ is increasing})$

$\therefore f'(x) = \frac{x}{3}$

$\therefore f'(6) = 2$ Ans.

6. Ans (0)

$f'(x) = 2x \int_1^{x^2} f(t)\,dt + 2x^3 f\left(x^2\right)$

$f'(1) = 2f(1)$

$f'(1) + f(1) = 3f(1) \ \& \ f(1) = 0$

7. Ans (1540)

$f(1) = 2$ and $f(x + y) = f(x) + f(y)$

$\Rightarrow f(x) = 2x$

$\sum_{r=1}^{100} (2r)^2 = 4\left[\frac{10.11.21}{6}\right] = 1540$

8. Ans (3)

$f(x) = \log_{\sqrt{5}}\left(\sqrt{2}(\sin x - \cos x) + 3\right)$ we know that

$-\sqrt{2} \le \sin x - \cos x \le \sqrt{2}, \quad \forall\, x \in R$

$[\text{since } -\sqrt{a^2+b^2} \le a \sin x + b \cos x \le \sqrt{a^2+b^2}]$

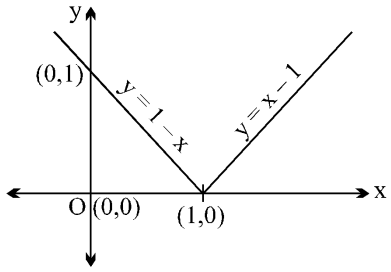
$\Rightarrow -\sqrt{2} \le (\sin x - \cos x) \le \sqrt{2}$

$\Rightarrow 1 \le \sqrt{2}(\sin x - \cos x) + 3 \le 5$

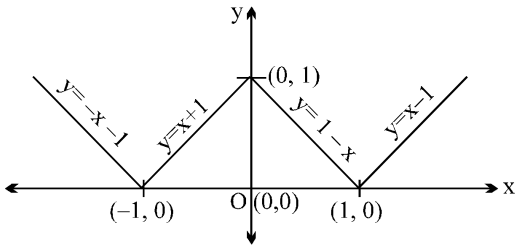
$\Rightarrow 0 \le \log_{\sqrt{5}}(\sqrt{2}(\sin x - \cos x) + 3) \le 2.$

Hence range is [0, 2]

9. Ans (5)



Graph of $y = f(x) = |x - 1|$



Graph of $y = f(|x|)$

Clearly the domain of $g(x) = \sin^{-1}(f(|x|))$ is $[-2, 2]$

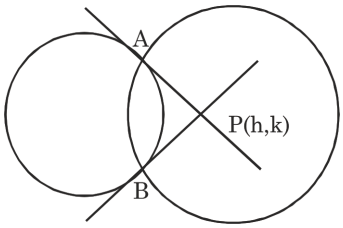
$\Rightarrow$ g (x) is non-differentiable at $x = -2, -1, 0, 1, 2.$

$\therefore$ Number of non-derivable points are 5. Ans.

10. Ans (2)

Equation of chord of contact w.r.t. P

$hx + ky = 1$



Equation of common chord is

$(\lambda - 3)x + (2\lambda + 2)y + 3 = 0$

$\Rightarrow \frac{\lambda - 3}{h} = \frac{2\lambda + 2}{k} = -3$

$\Rightarrow$ Equation of locus is $6x - 3y - 8 = 0$

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